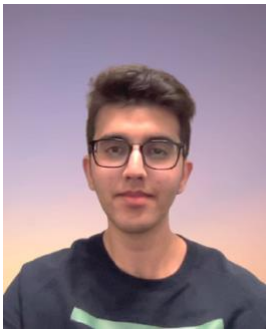

Aaryaman Sharma

Email: aaryaman1409@gmail.com
Mobile: +44 07984946199
LinkedIn: Aaryaman Sharma
GitHub: Aaryaman-1409

Education

University College London (UCL)
September 2021 - June 2024

Degree: Computer Science, BSc

Doha British School
2014 - 2021

International Baccalaureate: 44 points (math, biology, chemistry, English, ITGS, French)

IGCSEs: 9 Grade 9/A** and 1 Grade 8/A* (biology, chemistry, physics, mathematics, English literature, English language, French, geography, computer science & business)

Skills

Experienced with Python, Java, C, HTML, CSS, Arduino

Familiar with Swift, Haskell, SQL, JavaScript

Projects

- Designed and prototyped control system to maintain optimal conditions in bioreactor. Used ESP32 microcontroller and Arduino Uno, along with the Thingsboard IoT platform, to enable real-time viewing, altering and logging of bioreactor conditions.

- Used Django with PostgreSQL to create a discrete math learning tool for first-year computer science students. Web app randomly generates permutations and asks the user to solve mathematical problems relating to them. Hosted on Heroku.

- Created a MacOS application in Swift to interface with the Music app and automatically change the desktop wallpaper based on the currently playing song. Uses K-means clustering to generate the appropriate background color and allows the user to modify properties of the wallpaper, such as the scale of the artwork, in real-time. Available here.

- Designed a note-taking web app in Java using JSPs and Tomcat. Allows users to create, edit, delete, sort and search notes.

- Created program to access user's liked songs using Spotify Web API and scrape lyrics to those songs from AZLyrics.com. Lyrics are automatically added to the metadata on matching local music files.

- Created a platformer game called "ReKindle" using GML2. Designed all the sprites and assets. Published on itch.io

Work Experience

Qatar Invents at Texas A&M University, Researcher
2019 – 2020

Part of an intensive STEM program at Texas A&M University created to help solve the water crisis in Qatar. Created working prototypes of hydroponics systems tailored to Qatar's environmental needs and space limitations.

Evolution Sports Academy, Assistant Basketball Coach
2019 - 2021

Responsible for coaching a group of 15+ 9-12 year-olds. Created and demonstrated drills. Taught basketball strategy and acted as a secondary coach during games. Helped 2 kids move to an advanced class.

Priya's Cookbook, Web Developer
2019-2020

Worked with a client to create a fully-featured recipe website. Designed responsive layout, logos. Coded in JavaScript & HTML from scratch to accommodate client's needs. Features over 30 of the client's favorite recipes.

Volunteer Experience

Downtown Doha, Co-Founder & Lead Translator
2019- PRESENT

Student-led organization to provide essential resources to migrant construction workers in Qatar. Raised over 10,000 Qatari Riyals. Assembled donation packages for over 100 workers based on their needs. Arranged multiple visits to the worker living areas to distribute resources throughout the year. Received 2nd place in the Global Acts with Impact Award at The Hague International Model United Nations conference for our contributions.

Kavre Deaf School Nepal, Volunteer
2019-2020

Fundraised and donated over 3000 Qatari Riyals to school management. Responsible for plastering, cleaning and painting 5 classrooms and boarding rooms. Learnt basic Nepalese Sign Language. Helped teach 30+ children written English at a 5th grade level.

Awards

2020: Winner of prestigious Carnegie-Mellon Qatar Pi Day Math competition
2020: 2nd place winner- The Hague International Model United Nations Global Acts with Impact Award
2019: Pearson International GCSE High Achiever's Award for exceptional GCSE grades
2018: Winner of Qatar National Scratch Coding Competition